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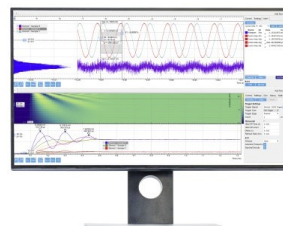
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Flow Characteristics of Bounded Self-organized Dust Vortex in a Complex Plasma

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Abstract. Dust clouds are often formed in many dusty plasma experiments, when micron size dust particles introduced in the plasma are confined by spatial non-uniformities of the potential. These formations show self-organized patterns like vortex or circulation flows. Steady-state equilibrium dynamics of such dust clouds is analyzed by 2D hydrodynamics for varying Reynolds number, Re , when the cloud is confined in an azimuthally symmetric cylindrical setup by an effective potential and is in a dynamic equilibrium with an unbounded sheared plasma flow. The nonconservative forcing due to ion flow shear generates finite vorticity in the confined dust clouds. In the linear limit ($Re \ll 1$), the collective flow is characterized by a single symmetric and elongated vortex with scales correlating with the driving field and those generated by friction with the boundaries. However in the high Re limit, ($Re \geq 1$), the nonlinear inertial transport ($\mathbf{u} \cdot \nabla \mathbf{u}$) is effective and the vortex structure is characterized by an asymmetric equilibrium and emergence of a circular core region with uniform vorticity, over which the viscous stress is negligible. The core domain is surrounded by a virtual boundary of highly convective flow followed by thin shear layers filled with low-velocity co- and counter-rotating vortices, enabling the smooth matching with external boundary conditions. In linear regime, the effective boundary layer thickness is recovered to scale with the dust kinematic viscosity as $\Delta r \approx \mu^{1/3}$ and is modified as $\Delta r \approx (\mu L_{\parallel}/u)^{1/2}$ in the nonlinear regime through a critical kinematic viscosity μ^* that signifies a structural bifurcation of the flow field solutions. The flow characteristics recovered are relevant to many microscopic biological processes at lower Re , as well as gigantic vortex flows such as Jovian great red spot and white ovals at higher Re .

INTRODUCTION

The dust clouds electro-statically suspended in a quasineutral plasma show characteristics of various phases of matter, ranging from crystalline to fluids, depending on the parameter of coupling between the dust grains ($\Gamma = \text{Potential Energy / Kinetic Energy}$) and the screening parameter ($\kappa = d/\lambda_D$) which is the ratio of average inter grain separation to the plasma Debye length [1–4]. Dusty plasmas with coupling parameters of $10 \leq \Gamma \leq 100$ over a range of fixed screening parameter κ behave like complex fluids [4–6]. When confined by an effective potential and subjected to external drivers of fluid flow, the suspended dust medium exhibits various self organized flow patterns like vortex or circulation, waves and shear layer formation apart from collective dust excitations extensively studied in the existing dusty plasma literature. [7–13]. The mechanisms underlying the vortex motion in these experiments include a range of possibilities that result into non-conservative forces on the suspended dust, either from its direct or indirect interaction with the surrounding plasma [7, 9, 14, 15]. The dynamics causing the dust vortex flows in experiments therefore remains complicated. These collective patterns in steady-state equilibria of the dust fluid represent a large category of driven-dissipative complex dynamical systems that continuously operate away from their thermodynamic equilibrium [16, 17]. The flow dynamics of dusty plasmas is therefore an attractive and accessible means for study of many complex flow system of natural flow systems including low Reynolds number biological flow systems, colloidal dynamics [18–20], upto very high Reynolds number large scale astronomical systems [21, 22]. Whether the interesting dust flow dynamics observed in a variety of recent dusty plasma experiments, often accessible by velocimetric techniques [12, 13], can be employed to study complex non-equilibrium thermodynamic systems is subject to availability of a systematic formulation of the dusty plasma in its simplest fluid-like dynamical regime.

The focus of the present study is characterization of dynamical structure of dust vortex flows at arbitrary Reynolds number Re with an external sheared driving field in an uniform ($\nabla n_{i,n} = 0$) and isothermal ($\nabla T_{i,n} = 0$) plasma. Study of vortex structure and its characteristics in such a complex medium presents an attractive option for analyzing

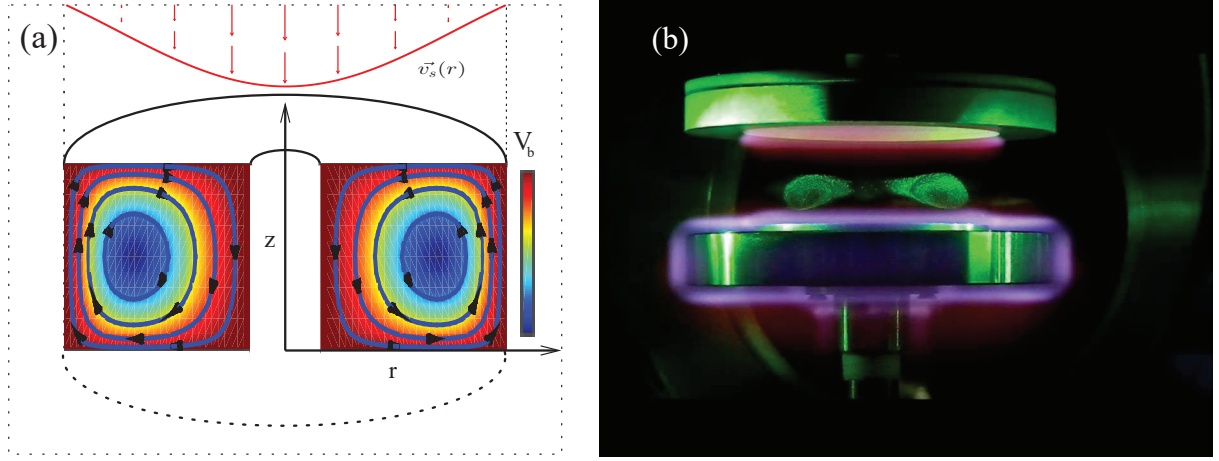


FIGURE 1. (a) Schematic representation of dust cloud confined toroidally by $V_b(r, z)$ and driven by shear flow field $v_s(r, z)$ through out the dust clouds. (b) The dusty plasma experimental setup by M. Kaur *et al.*, Phys. Plasmas **22**, 033703 (2015).

the dynamics of many relevant complex fluid flow in laboratory, industrial applications as well many natural flow processes. The rest of this paper (i) introduces the 2D hydrodynamic formulation and the cylindrical setup with axial and radial boundaries relevant to many existing experimental systems, (ii) presents solutions of the formulation and their variation over a range of Reynolds number values, (iii) characterizes the scaling of dust flow boundary layer thickness Δr with respect to the dust kinematic viscosity μ in a nonlinear, large Re limit recoverable beyond a critical dynamical transition identified as a structural bifurcation of dust flow field solutions with μ as bifurcation parameter.

SETUP AND 2-D HYDRODYNAMIC FORMULATION

A nearly incompressible dust cloud in its fluid-like regime is considered where the dust cloud is in a dynamical equilibrium with an unbounded background streaming plasma as described schematically in Fig. 1a. The interior of a highest potential contour confining the dust fluid is homogeneously filled with the dust fluid confined by an effective potential generated by the electrostatic and gravitational fields. The geometry of the confined dust fluid is considered identical to the recent experiments where a toroidal dust structure is formed having a poloidal dust flow in a glow discharge plasma as shown in Fig. 1b [11, 12]. In an azimuthally symmetric toroidal confined dust fluid recovered in the experiments, the dust fluid fills volume of the torus which is allowed to have an arbitrary shaped cross-section determined by the effective confining potential. However, for the computational simplicity this cross section of the torus is approximated in the present treatment as rectangular as illustrated schematically in Fig. 1a.

The effective confining potential $V(r, z)$ distribution is assumed within the boundaries of a finite section of an infinite cylinder of flowing plasma, or in the region where $0 < r/L_r < 1$ and $-1 < z/L_z < 1$, with symmetry along the azimuthal dimension $0 < \phi < 2\pi$ [23]. For the dust flow that follows incompressible and isothermal conditions, and has a finite viscosity, the dynamics is governed by the simplified Navier-Stokes equations in which the drive produced by the sheared ion drag and the friction produced by the stationary background neutral fluid can be suitably accounted as non-conservative momentum sources as follow [24],

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla P}{\rho} - \nabla V + \mu \nabla^2 \mathbf{u} - \xi(\mathbf{u} - \mathbf{v}) - \nu(\mathbf{u} - \mathbf{w}) \quad (1)$$

Here $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are the flow velocities of the dust, ions and neutral fluids, respectively. P and ρ are the pressure and mass density of the dust fluid, respectively, $V(r, z)$ is the total conservative potential for dust confinement, μ is kinematic viscosity, ξ is coefficient of ion drag acting on the dust and ν is the coefficient of friction generated by the stationary neutral fluid [25]. The overall combination of charged dust and background plasma is quasineutral and the electrons are in thermal equilibrium with the streaming ions and the confined dust for longer time scale. The incompressibility of the confined dust component is ensured by the expression $\nabla \cdot \mathbf{u} = 0$, which allows defining a streamfunction Ψ such that $\mathbf{u} = \nabla \times \Psi$ [24]. However, in case of an azimuthally symmetric flow, the dust motion can

suitably be treated in the 2-dimensional r - z plane such that the dust vorticity vector $\hat{\omega} = \nabla \times \mathbf{u}$ is directed purely along $\hat{\phi}$ and the stream function behave like a scalar potential for the two dimensional flows where $\mathbf{u} = \nabla \times (\psi \hat{\phi})$. The dust flow velocity components are thus expressible in term of ψ as, $u_r = -\frac{\partial \psi}{\partial z}$ and $u_z = \frac{1}{r} \frac{\partial(r\psi)}{\partial r}$ in the nonplanar flow system. In terms of the above definitions and the flow vorticity defined as $\omega = \nabla \times \mathbf{u}$, the model Eq. (1) for stationary neutrals ($\mathbf{w} = 0$), takes the form,

$$\nabla^2 \psi = -\omega, \quad (2)$$

$$(\mathbf{u} \cdot \nabla) \omega = \mu \nabla^2 \omega - (\xi + \nu) \omega + \xi \omega_s, \quad (3)$$

where $\omega_s = \nabla \times \mathbf{v}$ is the driver vorticity of the sheared flow of streaming plasma field $\mathbf{v}$. In real systems with alternate driving mechanisms, any non-zero $\nabla \times u_{i,n}$, $(\nabla Q_d \times \mathbf{E})$, $(\nabla u_{i,n} \times \nabla n_{i,n})$ and $(\nabla T_{i,n} \times \nabla n_{i,n})$ may contribute to ω_s . Solutions of the set of equations (2)-(3), with appropriate boundary conditions, represent the steady equilibrium flow of the confined dust fluid for any dynamical regime defined by the value of Reynolds number. The present analysis is based on the complete nonlinear numerical solutions of the system (2)-(3) obtained using a Successive Over-Relaxation (SOR) technique [26]. We now characterize the driven dust flow for a range of $\text{Re} \approx 10^{-2}$ to 10^2 , covering the relevant experimental and natural dust setups. The parameter set for the present solutions includes a known driving plasma field $\mathbf{v}$, streaming vertically downward (or along $-\hat{z}$, i.e., $v_r = 0$) and having a radial shear of the form of Bessel function representing the natural eigenmode of the present cylindrical setup, $v_z(r, z) = A_0 + A_1 J_0(\alpha_m \frac{r}{L_r})$. Here A_0 represents an offset and A_1 , the magnitude of radial variation of the unbounded plasma flow. The radial mode number α_m represents the zeros of corresponding ions vorticity coinciding with the external boundary location L_r .

CHARACTERIZATION OF DUST FLOW EQUILIBRIA

The boundary for solutions of (2)-(3) in the present treatment is defined by the effective potential that allows no dust flow across a confining potential well represented by one of the outer contour of the potential $V(r, z)$. The value of streamfunction ψ is fixed at this boundary while the solutions are obtained in the interior. This boundary value is defined depending on choice of the dust flow velocity and its variation along the boundary [27]. A choice of no flow across the boundaries ($u_\perp = 0$) is used for all the boundaries, and zero velocity along the boundaries ($u_\parallel = 0$) is imposed on them except at the boundary at $r = 0$ where the velocity is determined by the external driver. The complete flow solutions describing the driven dust dynamics in terms of 2D flow streamlines(contour or $r\psi$ in cylindrical rz cross section) and selected velocity u_z , u_r profiles are presented for small to high Reynolds number flow regimes in Fig. 2. The flow characteristics depend on various system parameters such as ion drag coefficient, neutral collision frequency, nature of boundaries, nature of driving field, and most importantly, the kinematic viscosity μ in the nonlinear flow regime [23, 27, 26].

The length and velocities in the present analysis are scaled to the system length or the domain dimension $L_0 \equiv L_r = L_z$ and shear ions flow strength U_0 , associated with vertically streaming ion velocity respectively. The parameters ξ , μ and ν accordingly have units derived for these scales. The entire analysis is done using a common combination of driver velocity field $v_z(r)$ and corresponding vorticity [26]. The solutions presented here correspond to a laboratory glow discharge argon plasma with micron size dust with parameters, $n \approx 10^9 \text{ cm}^{-3}$, $T_e \approx 3 \text{ eV}$, $T_i \approx 1 \text{ eV}$, where at the sheath entrance ions must stream with a value U_0 comparable to the ion acoustic velocity $c_s = \sqrt{T_e/m_i}$. For our parameters, the value of ion drag coefficient and neutral collision frequency can be estimated as $\xi \approx 10^{-4} U_0/L_0$ and $\nu \approx 10^{-3} U_0/L_0$, respectively [25]. For a typical system size, $L_0 \approx 10 \text{ cm}$, the range of kinematic viscosity μ can similarly be chosen to be $\mu \approx 6 \times 10^{-4} U_0 L_0$ which corresponds to small Reynolds numbers ($\text{Re} \approx 1$) of the dust flow consistent with the linear viscous regime [27].

The streamlines patterns at low Reynolds number regimes ($\text{Re} \leq 10$) show an increasing effect of the down-streaming driver flow which introduces increasing asymmetry in the streamline patterns with reducing μ as seen in Fig. 2a,b. With further reduction of μ ($\approx 10^{-5} U_0 L_0$ or $\text{Re} \approx 70$) plotted in Fig. 2c, the flow structure acquires larger asymmetry showing a primary circular region with almost uniform vorticity $\omega = \text{const.}$, surrounded by a region of highly shear upto confining boundaries. Moreover, at very high $\text{Re} \geq 100$, the core uniform vorticity region is further strengthened and additionally surrounded by a virtual boundary resulting from a boundary layer separation due to no-slip boundaries. The shear region near the corners and domain boundaries is filled with small scale vortices having co-rotating as well as counter rotating circulation, depending upon the dominant driver present in the region as shown in Fig. 2d. The small scale vortices are either driven by the external fields or the primary core vortex itself.

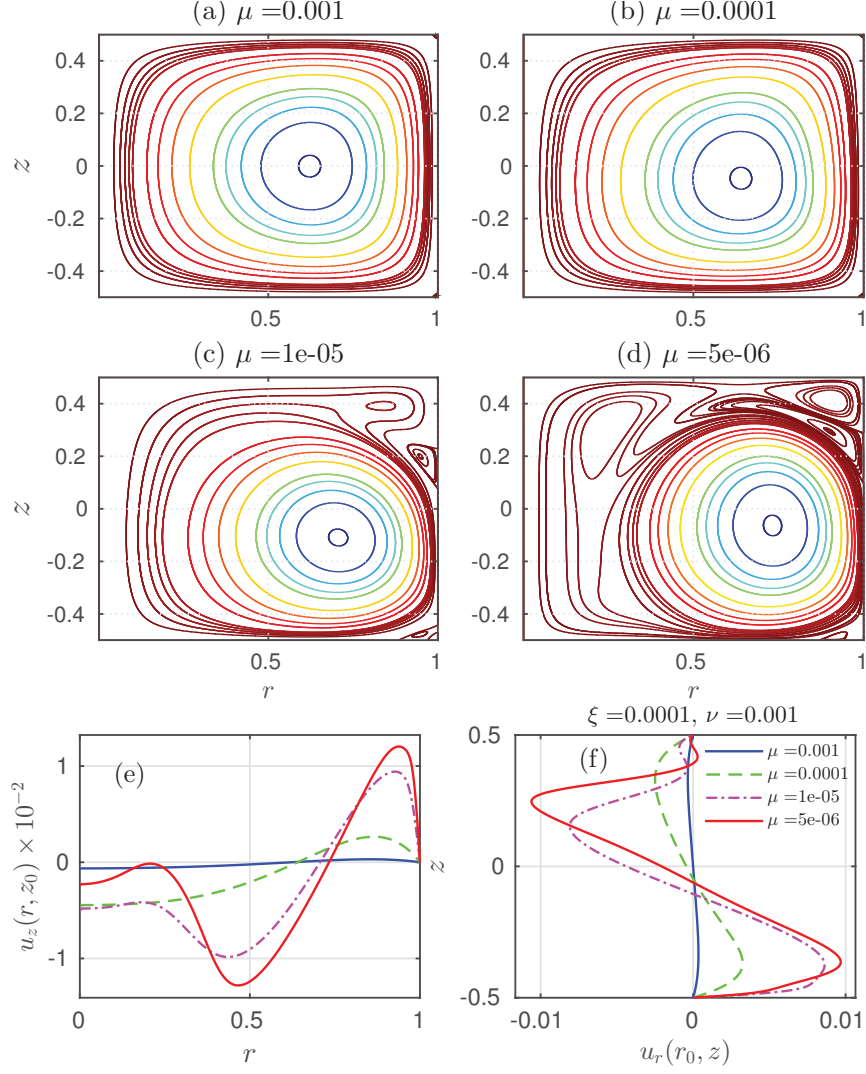


FIGURE 2. Dust flow streamline patterns and corresponding $u_z[U_0]$, $u_r[U_0]$ profile through center of circulation (r_0, z_0) for varying kinematic viscosity μ .

The series of structural changes with increasing Re is recovered involving transition to a new dynamic equilibrium which are attributed to nonlinear convective transport ($\mathbf{u} \cdot \nabla \mathbf{u}$) dominating over the diffusive processes. The sudden and spontaneous structural change to a new fluid flow pattern in nonlinear regime through a small changes in μ^* was already identified as caused by a structural bifurcation and presented in terms of boundary profile of vorticity. Specifically, the vorticity profile having a degenerate singular point bifurcates, at a μ^* into a set of two isolated points since the profile begins to intersect the axis $\omega = 0$ at two locations around the point of minimum value [26]. An additional impact of this bifurcation is presented here as observed in terms of downward directed dust flow velocity at $r = 0$ which shows as a function of μ and maxima at $\mu^* \approx 10^{-5} U_0 L_0$ as seen in Fig. 2e,f. As a clearly understood physical reason governing this effect we identify that the effect of momentum loss to the no slip boundaries (i.e., bottom, right and top boundaries) is allowed to be convected to the free boundary at $r = 0$ in this Boundary Layer Separated (BLS) regime, $\mu < \mu^*$. As an important effect, the maximum momentum transferred by the driver at $r = 0$ is getting convected to the boundaries via the outer streamlines, rather than getting diffused through the interior constant vorticity core. The uniformities of vorticity ω in the circular region is revealed by an almost uniform shear in the flow velocity, of both its components u_z and u_r , about its points of cross-section at the center of the circulation, $(r_0, z_0) \approx (0.7, 0.08)L_0$.

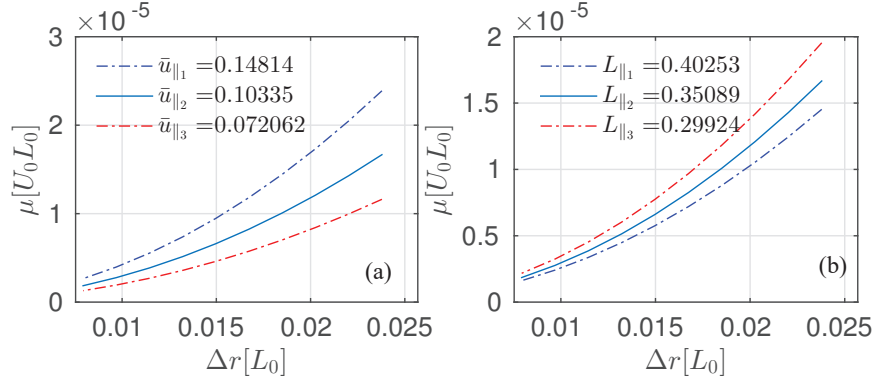


FIGURE 3. Characterization of scaling law for boundary layer thickness, $\Delta r \approx (\mu L_{\parallel}/u)^{1/2}$ in the nonlinear regime ($Re \gg 1$), beyond a critical kinematic viscosity μ^* .

We additionally characterize the nonlinear boundary layer thickness scaling law which is recovered to show a velocity dependence beyond this critical transition at μ^* . Note that a velocity independent effective boundary layer thickness was recovered as $\Delta r \approx \mu^{1/3}$ in the linear regime, ($Re \leq 1$) [27], which is then modified as $\Delta r \approx (\mu L_{\parallel}/u)^{1/2}$ in the nonlinear regime ($Re \gg 1$) [26], beyond a critical kinematic viscosity μ^* . This newly recovered dependence on the boundary layer flow velocity u and the characteristic velocity gradient scale along the streamlines $L_{\parallel}$ is characterized in in fig. 3(a) and (b), respectively. The dependence $\mu(\Delta r)$ presented for a range of u and $L_{\parallel}$ values should allows one to estimate the dust viscosity using the outputs of particle imaging velocimetry (PIV) technique which might provide local velocity and associated length scales in the flow system.

SUMMARY AND CONCLUSIONS

The flow characteristics of a confined dust fluid suspended in an unbounded streaming plasma are studied for a wide range of Reynolds number flow regimes using vorticity-streamfunction formulation of 2D hydrodynamic model. The confined and driven dust fluid exhibits vortex motion representing a dynamical model for many driven dissipative system operating away from their thermodynamic equilibrium. The nonlinear properties of the driven dissipative 2D dust vortex flow suspended in a plasma are studied in term of streamline patterns and corresponding flow velocimetric quantities for varying Reynolds numbers. In presence of 2D heterogeneous boundary conditions applied to dust flow in a nonplanar system, the major nonlinear effects cause the boundary layer to separate from the domain boundary and generate secondary vortices, adapting to the structure of the domain boundary interfacing the system surrounding. The impact of bifurcation on the dust flow structure, already noted in terms of vorticity profiles, is shown on the velocity at the center $r = 0$ where dust flow velocity shows a maximum with variation in μ . The newly obtained velocity field dependent nonlinear scaling is characterized in order to allow its possible validation in the experiments using the outputs of velocimetric techniques. The recovery of a uniform vorticity core below $\mu = \mu^*$ is illustrated from the profiles of flow velocity components plotted along the spatial dimensions r and z of the 2D setup. The solutions obtained in present analysis are relevant to recent observations in dusty plasmas setups where dust flow is recoverable with considerably uniform vorticity in a limited spatial domain. The recovered flow characteristics are also relevant to many natural flows such as microscopic biological activity at lower Re , and large scale highly stable vortex flows with uniform vorticity core, such as Jovian great red spot and white ovals at higher Re [21, 22].

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